AI-1 Spring 26 Project Proposal

Team Members

Section No	Name in ARABIC
1	عبدالرحمن محمد ابراهيم محمد
4	عبدالرحمن محسن محمد
8	مصطفى محمود مصطفى مكرم

Team ID ( )

Idea Short Name (2 D car game)

Idea Description

2D classic Car game like subway on mobile phones
The car should follow traffic rules and not crashing each other

Agent Architecture

☐ Simple Reflex Agent

√ Model-Based Reflex Agent

√ Goal-Based Agent

√ Utility-Based Agent

Problem-Solving Framework

√ Classical Search

√ Local Search (Optimization)

√ Adversarial Search (Two-Player Games)

√ Constraint Satisfaction Problem

Proposed Algorithm(s)

Heuristic Evaluation **Hill Climbing** **Minimax + Alpha-Beta Pruning** **Genetic Algorithm**

PEAS

Performance Measures	**Score**+1 for every NPC car successfully passed **Survival time** How long the player/AI stays alive **Best score** Tracked across sessions (best_score) **Crash avoidance** Penalized by game over on collision **Traffic law obedience** Penalized by game over for running red light**Speed maintenance** Rewarded for staying near max speed without crashing
Environment	**Road** — **4-lane road, 800×700 pixels** **NPC vehicles** — Cars, trucks, vans, ambulances spawning from top **Traffic lights** — Red / Yellow / Green with timed cycles, spawning from top **Lane boundaries** — Hard walls at left and right edges **Game modes** — Normal, Speedy, Emergency — different speeds & densities **GA-evolved difficulty** — Spawn rate, speed bonus, vehicle count evolve each round
Actuators	**Move Left** **Move Right** **Brake (Hard)** **Auto** **Accelerate**
Sensors	Lane detector Vehicle proximity Traffic light state Traffic light prediction Speed sensor

	Collision detector Lane safety scores Keyboard input

Environment Properties

☐ Fully Observable √ Partially Observable	☐ Deterministic √ Stochastic √ Strategic	√ Discrete √ Continuous
☐ Single-Agent √ Multi-Agent	√ Sequential ☐ Episodic	☐ Static √ Dynamic